

BIG O ALGORITHM ANALYSIS

If your data is already in sorted order, performing a binary search is much faster than a *linear search*, which starts from one end of an array and checks each value in a brute-force manner. We can compare the efficiency of these two algorithms with *big O algorithm analysis*. “Further Reading” at the end of this chapter has links to more information on this topic.

If your data isn't in sorted order, sorting it first (using an algorithm such as quicksort or merge sort) and then performing a binary search would be slower than just performing a linear search. However, if you need to perform repeat searches on the data, the performance gain of binary search pays for the sorting. It's like spending an hour to sharpen an axe before cutting down trees; the increased speed of chopping with a sharp axe makes up for the hour spent sharpening it.